

Retail Industry

The retail industry is formed by all companies that sell goods and services directly to consumers. It includes both physical stores, like groceries, convenience, discounts, department stores, and online platforms. The business model is simple: retailers buy large quantities of products from manufacturers and then sell them in smaller quantities to end consumers. This industry matters because it gives people easy access to their daily needs, no matter which channel they choose to shop through.

AI personalization in eCommerce

AI personalization consists of using Artificial Intelligence technologies to provide customized shopping experiences. These systems analyze the individual user's shopping behavior, preferences, and data patterns, and adapt in real-time to provide highly relevant content, offers, and product suggestions.

These systems work in four steps:

1. Data collection: the system collects data like browsing history, cart activity, and previous purchases.
2. The system analyzes this data using machine learning algorithms to detect patterns and predict user preferences.
3. The AI groups customers into dynamic segments based on these patterns.
4. Based on its predictions, the platform changes what content, offers, or products the user sees.

AI personalization has shown to increase revenue per user by 166%. These types of systems not only increase sales, they also enhance customer loyalty by 31% and make customers feel acknowledged and appreciated.

AI Inventory Management

Inventory planning is hard to scale. It's not just about counting products. Demand changes daily and depends on factors like weather, holidays, local events, and supply delays.

Moreover, it changes across different locations: a product that sells fast in one city can barely move in another.

To solve this problem, an AI model analyzes massive amounts of data, including past sales, seasonal patterns, pricing changes, and weather, to predict demand ahead of time.

The impact of AI-powered inventory management is clear. Walmart, for example, uses a neural network built specifically to predict demand at the store, allowing it to plan inventory

more precisely. This results in fewer out-of-stock situations and better product availability, giving customers a smoother shopping experience and letting employees spend more time helping customers instead of managing stock.

Job displacement

AI is automating a lot of processes in the retail industry, like inventory management, marketing planning, and even the cashier job. However, the industry is now experiencing a backlash against self-checkout and cashier less systems like Amazon's "Just Walk Out," with many retailers going back to classic human cashiers.

Dollar General has eliminated self-checkout in about 12,000 stores, and Five Below is working to remove it completely in its highest-risk locations. This new trend of eliminating self-checkout technology, powered by AI or not, comes down to two main reasons. The first is that it reduces theft and loses inventory. The second is pursuing a better customer experience, since a lot of customers complain that the machines are slow, confusing, or simply feel impersonal.

Privacy and data use

This is a fundamental aspect of ethical AI and its correct implementation in retail. The industry collects data from loyalty programs, location tracking, receipts, browsing behavior, and third-party sources. To protect customer privacy, retailers must ensure that this data is collected, stored, and processed securely.

This means asking for explicit consent before collecting and using data, using that data only for its declared and legitimate purpose, collecting only what is necessary, and controlling who has access to it. A real example of what happens when companies fail to do this is Clearview AI, which scraped billions of facial images from the internet without consent to build a facial recognition database. The company ended up paying a \$51.75 million settlement under Illinois' Biometric Information Privacy Act.

This matters to companies not just ethically, but financially. Any data breach can result in loss of customer trust, reputational damage, and serious legal consequences.

Future trends

The future of retail with AI is heading toward a much more personalized and contextualized shopping experience than what we have today. In the future, this contextual customization won't be limited to digital platforms only. We will probably be able to walk into a physical store and receive real-time feedback based on the data companies already have about us, like our preferences or shopping history.

A good example of this would be immersive stores, where you could simply point out your phone at a product, and an AI agent would tell you about it, or show you a demonstration of how it works or how to use it. This kind of experience would make shopping faster and more informative, without needing to ask a store employee.

In fact, some of these changes are already happening right now, like inventory management improvement. Better inventory management gives customers a smoother experience, since it helps avoid situations where a product is out of stock.

In conclusion, AI is already revolutionizing the retail industry, and we are starting to see a much more personalized shopping experience, both in physical stores, online stores, and even on social media. This is especially true on platforms like TikTok, where the algorithm, which is essentially AI, shows you exactly the product you need, want to buy, or have been thinking about.

This is an industry I wouldn't want to work in. There's no real specific reason; it just doesn't interest me that much.

Citation:

Assosia. (2024, March 8). Retail Industry Overview | Key trends and market analysis.

<https://www.assosia.com/sectors-channels/retail>

EComposer. (2026, January 29). How AI personalization is transforming eCommerce in 2026.

<https://ecomposer.io/blogs/ecommerce/ai-personalization-ecommerce>

Das, H. (2026, February 24). Walmart's AI Inventory Optimization: How Data Keeps Shelves

Full. Xaltius Academy. <https://xaltiusacademy.com/walmarts-ai-inventory->

[optimization-how-data-keeps-shelves-full/](https://xaltiusacademy.com/walmarts-ai-inventory-optimization-how-data-keeps-shelves-full/)

Some retailers are pulling back from self-checkout. (2024, July 4). [Video]. NBC News.

<https://www.nbcnews.com/business/business-news/major-retailers-are->

[backtracking-self-checkout-rcna160234](https://www.nbcnews.com/business/business-news/major-retailers-are-backtracking-self-checkout-rcna160234)

Gonzalez, A. I. G. (n.d.). Ethical Considerations of AI in Retail: Best practices for Responsible

AI. Hexaware Technologies. <https://hexaware.com/blogs/ethical-considerations-of->

[ai-in-retail-best-practices-for-responsible-ai/](https://hexaware.com/blogs/ethical-considerations-of-ai-in-retail-best-practices-for-responsible-ai/)

Team, T. P. L. S. a. G., Navetta, D., Geiser, L., Waltz, D., Gouverneur, N., Team, T. P. L. S. a. G.,

Navetta, D., Geiser, L., Waltz, D., Gouverneur, N., Team, T. P. L. S. a. G., Navetta, D.,

Geiser, L., Waltz, D., & Gouverneur, N. (2026, May 5). \$51.75M Settlement in

Clearview AI Biometric Privacy Litigation Illustrates Creative Resolution for Startups

Facing Parallel Litigation and Enforcement Action. Regulatory Oversight.

<https://www.regulatoryoversight.com/2025/04/51-75m-settlement-in-clearview-ai->

[biometric-privacy-litigation-illustrates-creative-resolution-for-startups-facing-](https://www.regulatoryoversight.com/2025/04/51-75m-settlement-in-clearview-ai-biometric-privacy-litigation-illustrates-creative-resolution-for-startups-facing-)

[parallel-litigation-and-enforcement-action/](https://www.regulatoryoversight.com/2025/04/51-75m-settlement-in-clearview-ai-biometric-privacy-litigation-illustrates-creative-resolution-for-startups-facing-parallel-litigation-and-enforcement-action/)

COSIN CONSULTING S.L. (2025, May 9). *The future of artificial intelligence in retail*.

Cosin Consulting. <https://cosinconsulting.com/en/the-future-of-artificial-intelligence-in-retail/>